ABHISHEK SHARMA

Senior AI Data Engineer | Cloud Data Platforms & Production GenAI Systems

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PROFESSIONAL SUMMARY

AWS & Databricks Certified Senior Data & AI Engineer with 3+ years of enterprise experience architecting cloud Lakehouses (**PySpark, Apache Iceberg, AWS Glue, Athena, Lake Formation**) and production GenAI systems (**RAG, PostgreSQL pgvector, Model Context Protocol [MCP], multi-agent orchestration, and autonomous self-healing agents**). Built high-scale data platforms processing **15M+ daily events (1.2+ TB)** and deployed sub-20ms p99 semantic caching layers across enterprise telemetry streams. Recipient of two executive commendations (EchoStar CEO Hamid Akhavan and Inadev COO Abhishek Goel) for engineering a Gemini 2.5 Flash code compiler delivering multi-million-dollar ETL licensing cost reductions and rapid enterprise reporting. Proven track record in slashing analytical query latency by **65%**, reducing LLM inference costs by **72%**, and enforcing zero-trust HIPAA/GxP compliance.

CORE COMPETENCIES

Cloud Lakehouse & Big Data Platforms: AWS (Glue, Athena, Iceberg, Lake Formation, S3, Lambda, Step Functions, RDS), Databricks, GCP (BigQuery, Cloud Storage), Microsoft Fabric, PySpark, Python, SQL, Medallion Architecture (Bronze/Silver/Gold), ETL/ELT Design, Z-Order Partitioning, Table Compaction.

GenAI, RAG & Agentic Systems: Retrieval-Augmented Generation (RAG), PostgreSQL pgvector (HNSW Indexing, Dense+Sparse Hybrid Search, Reciprocal Rank Fusion), Model Context Protocol (MCP), Multi-Agent State Graphs, AWS Bedrock (Claude 3.5 Haiku/Sonnet), Gemini 2.5 Flash, Azure OpenAI, Prompt Caching, Semantic Routing, Ragas Hallucination Guardrails, LLMOps.

Backend, Observability & Telemetry: FastAPI, TypeScript Fastify, Redis/BullMQ, Docker, CI/CD (GitHub Actions Quality Gates), Playwright, Prometheus, Grafana, Loki, Grafana Tempo (Distributed Tracing), OpenTelemetry (OTel).

Data Governance & Zero-Trust Security: AWS Lake Formation (Tag-Based Access Control [LF-TBAC], Dynamic Column PII Masking), CDGC Data Cataloging, HIPAA/GxP Regulatory Compliance, OAuth2/Azure AD Multi-Tenant SSO, PostgreSQL Row-Level Security (RLS).

KEY EXECUTIVE RECOGNITIONS

- **Executive Commendation — CEO Hamid Akhavan (EchoStar / Hughes Network Systems):** Recognized for independently architecting a Gemini 2.5 Flash GenAI code compiler that automatically parsed complex legacy Informatica XML mapping DAGs into optimized PySpark and BigQuery SQL, reducing migration cycle times by **85%** and eliminating enterprise ETL vendor licensing costs.
- **Executive Commendation — COO Abhishek Goel (Inadev India):** Awarded for rapid end-to-end delivery on the Gartner Managed Networks project, architecting and deploying 39 operational BigQuery and SAP WebI reports in under 3 weeks (4x faster than standard delivery timelines).

PROFESSIONAL EXPERIENCE

Inadev India Pvt Ltd

Kolkata, WB | Oct 2023 – Present (3+ Years)

Accelerated Career Progression: Associate Data Engineer → Data Engineer (Cloud Lakehouse) → Lead / Senior AI Data Engineer

Lead / Senior AI Data Engineer

May 2026 – Present

DevOps AIris — Enterprise AIOps Incident Orchestration & pgvector Platform

- Engineered a high-throughput incident triage backend delivering **99.2% retrieval precision** by architecting FastAPI microservices with PostgreSQL Row-Level Security (RLS) and **pgvector hybrid search** (HNSW dense embeddings + BM25 sparse reciprocal rank fusion).
- Reduced recurring incident triage overhead and MTTR by **45%** with **sub-20ms p99 query latency** by engineering an in-memory semantic solution-caching layer over unified Prometheus metrics, Loki log streams, and Grafana Tempo distributed traces.
- Automated real-time incident ticketing and post-mortem operational runbook generation by developing custom **Model Context Protocol (MCP) servers** establishing bidirectional synchronization between Keep UI, Jira Cloud, and Confluence.

AutoFlow Studio — Autonomous Web Testing & Self-Healing Agent Platform

- Automated resilient E2E web-testing and dynamic selector remediation across 28 relational database schemas, slashing manual test maintenance overhead by **80%** using TypeScript Fastify, Redis/BullMQ task queues, and Playwright execution workers.
- Decreased LLM inference operating costs by **72%** with sub-second decision latency by deploying stateful multi-agent execution graphs with Anthropic prompt caching and semantic model routing (tiering Claude 3.5 Haiku for DOM diffing and Claude 3.5 Sonnet for failure recovery).

Welo Data AI — Agentic Reinforcement-Learning Simulation Platform

- Engineered high-concurrency simulation environments to benchmark and train autonomous AI agents across **200+ specialized enterprise operational roles**.

Data Engineer — Cloud Lakehouse & Healthcare Pipelines
Nov 2025 – Apr 2026

Quantum Research — Clinical Trial Healthcare Data Platform

- Accelerated analytical query speeds by **65%** and eliminated concurrency locks across **15M+ daily clinical-trial events (1.2+ TB raw data)** by building event-driven AWS Step Functions and PySpark Glue pipelines writing to an **Apache Iceberg** Lakehouse with automated metadata table compaction and Z-order partitioning.
- Enforced zero-leakage patient confidentiality passing **100% of HIPAA and GxP regulatory audit gates** by implementing **AWS Lake Formation tag-based access control (LF-TBAC)** and automated dynamic column PII masking at the storage boundary.
- Guaranteed zero schema-drift production outages across ETL releases by authoring GitHub Actions CI/CD quality gates enforcing automated PySpark data contracts, bytecode linting, and schema validation before deployment.

Associate Data Engineer / Analyst
Oct 2023 – Nov 2025

Hughes Network Systems / EchoStar — Global Satellite Communications

- Eliminated multi-million-dollar proprietary ETL licensing constraints by architecting a **Gemini 2.5 Flash GenAI compiler** that parsed legacy Informatica XML maps into optimized BigQuery SQL/PySpark jobs, earning the EchoStar CEO commendation.
- Led Oracle-to-BigQuery cloud migrations using Informatica IICS; built high-integrity Bronze/Silver/Gold Lakehouse tables with CDGC metadata governance and automated reconciliation checks.
- Engineered and deployed 39 Gartner Managed Networks analytics reports in under 3 weeks (BigQuery + SAP WebI), earning COO-level recognition.
- Developed a secure Streamlit-based internal production support chatbot with role-based access control and telemetry logging for rapid error troubleshooting.

FEATURED PROJECT — AI HACKATHON

InsureVoice — Voice-Driven Multi-Agent AI Insurance Advisor Google Cloud Rapid Agent Hackathon | Jun 2026

- Co-built a voice-driven multi-agent AI insurance platform using **Google Agent Development Kit (ADK)** and **Gemini 2.5 Flash Lite** that converts conversational customer inquiries into explainable, compliant Top-3 policy recommendations.
- Integrated **Elastic ELSER v2 semantic search via Model Context Protocol (MCP)** to retrieve regulatory policy documents with sub-50ms deterministic orchestration and sub-200ms voice synthesis via Google Chirp 3 HD TTS.

CERTIFICATIONS

- **Databricks Certified Data Engineer Associate** — Databricks Academy (Feb 2026 – Feb 2028)
- **AWS Certified Cloud Practitioner** — Amazon Web Services (Apr 2023 – Mar 2029)
- **Anthropic AI Fluency: Framework & Foundations** — Anthropic (Apr 2026)
- **Career Essentials in Generative AI** — Microsoft & LinkedIn (Oct 2026)
- **AI for All** — Intel & Skill India Digital Hub (Oct 2026)
- **Generative AI Fundamentals & Databricks Fundamentals Accreditation** — Databricks Academy (Sep 2025)
- **AI/BI for Data Analyst** — Databricks Academy (Oct 2025)

EDUCATION

Bachelor of Technology (B.Tech) in Electronics & Communication Engineering Aug 2019 – Jul 2023
Swami Vivekananda Institute of Science and Technology (MAKAUT, West Bengal) CGPA: 9.14 / 10.0